A Gaussian Splatting Pipeline for Fraud-Resistant 3D Vehicle Reconstruction in Sri Lankan Motor Insurance

***Abstract*—Motor vehicle accident claim processing in Sri Lanka relies predominantly on two-dimensional photographic submissions, exposing insurers to claim recycling fraud, inaccurate damage assessment, and failure under low-light conditions. This paper presents a serverless Gaussian Splatting pipeline that converts smartphone photographs into photo-realistic, interactive 3D vehicle models, enabling insurance officers to rotate, pan, and inspect volumetric damage from any angle directly in a web browser without specialist hardware. The pipeline integrates a three-tier adaptive brightness detection system, Zero-Reference Deep Curve Estimation for low-light enhancement, Structure-from-Motion for camera pose estimation, and neural Gaussian Splatting for per-scene 3D model training on an NVIDIA L4 GPU. When mean image brightness falls below 50 out of 255, the pipeline delivers enhanced photographs rather than attempting a failed reconstruction, ensuring a useful output under all evaluated lighting conditions. A multi-layer fraud verification framework cross-references insurance call metadata, per-image GPS coordinates, capture timestamps, and third-party identity documents to anchor each claim to a verified time and location. Evaluated across over 30 real pipeline executions on an NVIDIA L4 GPU, the system achieved end-to-end reconstruction times of 20.3 and 22.4 minutes for well-lit submissions, brightness restoration of 90.1 points for critically dark inputs, a BRISQUE quality improvement from 47.81 to 43.22 following neural enhancement, correct intelligent degradation triggering confirmed across all critically dark submissions, and reconstruction quality of 21.78 dB PSNR with SSIM of 0.6536. To the best of the authors' knowledge, no prior work has applied Gaussian Splatting to motor insurance claim documentation in the Sri Lankan context.**

***Index Terms*—Gaussian Splatting, 3D Vehicle Reconstruction, Insurance Fraud Detection, Zero-DCE, Low-Light Enhancement, Structure-from-Motion, COLMAP, nerfstudio, Neural Rendering, Sri Lanka Motor Insurance**

I. Introduction

The motor insurance industry in Sri Lanka processes accident claims through a workflow that has remained largely unchanged despite increasing smartphone penetration. Claimants submit photographs via messaging platforms or basic web portals, and insurance officers manually assess damage from these two-dimensional images. This approach carries three fundamental technical failures that undermine claim integrity.

First, two-dimensional photographs provide no depth information, making it impossible to verify the volumetric extent of vehicle damage remotely. A photograph of a dented panel cannot convey whether the deformation is 2mm or 20mm deep, a distinction that may determine repair cost by an order of magnitude. Second, existing systems provide no mechanism to prevent claim recycling, where a claimant submits photographs of historical damage as evidence of a new incident. Without spatial and temporal verification, insurers cannot distinguish a genuine claim from a fraudulent resubmission. Third, a significant proportion of road accidents occur at night or in low-light conditions. Photographs captured in such environments are frequently too dark for reliable damage assessment, forcing insurance officers to dispatch manual inspectors, a process that delays settlement and increases operational cost.

These failures are well-documented in the broader insurance technology literature. Gupta et al. [1] survey AI and computer vision methods for vehicle damage assessment, highlighting the limitations of flat-image approaches and the potential of multi-view analysis. Somnuk and Kanchanasut [2] demonstrate that 3D reconstruction from multiple 2D photographs is technically feasible for accident scenes, while Kholodkov et al. [3] apply photogrammetry to reconstruct accident sites with spatial fidelity. Schrijver et al. [4] identify claim recycling as a primary vector of insurance fraud. Despite this body of work, no existing system integrates 3D reconstruction, adaptive low-light enhancement, and intelligent degradation into a single deployable insurance claim pipeline.

This paper presents an end-to-end pipeline that addresses all three failures. The system accepts unordered smartphone photographs uploaded at the accident scene, applies adaptive brightness correction, estimates camera poses via COLMAP [5], trains a per-scene Gaussian Splat using nerfstudio splatfacto [6] on a serverless NVIDIA L4 GPU via Modal.com, and streams the resulting `splat.ply` to a WebGL viewer in the insurance officer dashboard.

The primary contributions of this paper are as follows:

- A complete serverless Gaussian Splatting pipeline for motor accident claim documentation, running on an NVIDIA L4 GPU via Modal.com with zero idle cost.
- A three-tier adaptive brightness detection system that routes each submission to the appropriate enhancement and reconstruction path.
- A custom three-stage low-light enhancement chain combining Zero-DCE [7] neural tone curve estimation, a piecewise shadow lift lookup table, and YCbCr-space noise reduction.
- An intelligent degradation mechanism ensuring a useful output under all lighting conditions including critically dark accident scenes.

- Empirical evaluation across over 30 real pipeline executions demonstrating reconstruction in 16.8–22.4 minutes, brightness restoration of $+90.1$ points for critically dark inputs, and reconstruction quality of PSNR 21.78 dB and SSIM 0.6536.
- A multi-layer fraud verification framework that cross-references insurance notification call metadata, guided capture timestamps, per-image GPS coordinates, and third-party identity documentation ensuring spatial and temporal consistency across all claim evidence.

The remainder of this paper is organised as follows. Section II reviews related work. Section III describes the system architecture. Section IV details the low-light enhancement chain. Section V presents the reconstruction methodology. Section VI reports experimental results. Section VII discusses limitations and future work. Section VIII concludes.

II. Literature Review

A. 3D Accident Reconstruction

Somnuk and Kanchanasut [2] proposed a convolutional neural network approach to reconstruct vehicle accidents from multiple 2D images, demonstrating that spatial relationships between vehicles can be recovered from standard photographs. Kholodkov et al. [3] applied photogrammetry and spatial modelling to reconstruct accident sites at a level of fidelity superior to traditional photo-based documentation. Gupta et al. [1] survey the broader landscape of AI and computer vision methods for vehicle damage assessment, concluding that multi-view analysis is necessary for accurate volumetric damage estimation. Collectively, these studies establish the technical feasibility of 3D reconstruction for insurance contexts, but none address low-light failure modes or the intelligent degradation problem.

B. Low-Light Image Enhancement

Zero-DCE [7], published by researchers at the Chinese University of Hong Kong, is a lightweight convolutional neural network that enhances dark images without requiring paired bright/dark training data. Unlike traditional histogram equalisation, Zero-DCE predicts per-pixel tone curves applied iteratively across 8 passes, lifting darker regions selectively while leaving highlights intact. Its zero-reference property makes it directly applicable to accident scene photography where no curated paired dataset exists. Existing photogrammetry pipelines do not integrate neural enhancement as a preprocessing stage, leaving low-light reconstruction as an unaddressed failure mode.

C. Structure-from-Motion and Neural Rendering

COLMAP [5], developed at ETH Zurich, is the established open-source library for Structure-from-Motion, recovering 3D scene geometry from unordered 2D photographs via SIFT feature extraction [8], exhaustive feature matching, and bundle adjustment. Kerbl et al. [9] introduced 3D Gaussian Splatting, representing scenes as millions of Gaussian primitives optimised via gradient descent to match input photographs from every angle. Unlike traditional polygon mesh outputs, Gaussian Splats support real-time interactive rendering in a browser without specialist hardware, making them uniquely suited to insurance officer dashboards. Gaussian Splatting [9] builds on the neural radiance field paradigm of Mildenhall et al. [10]. Nerfstudio [6] provides the splatfacto training framework used in this work.

D. Insurance Fraud Detection

Schrijver et al. [4] conducted a systematic literature review of automobile insurance fraud detection methods, identifying claim recycling as a primary fraud vector. Data mining and anomaly detection approaches are effective at identifying suspicious claim metadata patterns, but cannot verify the spatial authenticity of submitted photographs. Volumetric 3D reconstruction directly addresses this gap: a Gaussian Splat model generated from real accident photographs encodes depth and geometry that cannot be fabricated from pre-existing flat images.

E. Research Gap

Table I confirms that no existing system simultaneously provides neural low-light enhancement, Gaussian Splatting reconstruction, intelligent degradation for critically dark inputs, and browser-based interactive viewing in a single insurance claim pipeline. The proposed system addresses all four dimensions.

TABLE I
Capability Comparison of Existing Systems vs. Proposed System

Capability	**[2]**	**[3]**	**[4]**	**Ours**
Neural low-light enhancement	×	×	×	✓
3D volumetric reconstruction	✓	✓	×	✓
Intelligent dark-scene degradation	×	×	×	✓
Browser interactive viewer	×	×	×	✓
Insurance claim integration	×	×	✓	✓
Serverless GPU provisioning	×	×	×	✓

III. System Architecture

A. System Overview

The proposed system connects a React Native mobile application to an insurance officer web dashboard through a serverless GPU reconstruction pipeline. When a claim arrives, the insurer triggers 3D model generation directly from the dashboard. The high-level data flow is:

$$\textit{Mobile App} \rightarrow \textit{Cloudflare R2} \rightarrow \textit{FastAPI} \rightarrow \textit{Modal GPU} \rightarrow \textit{R2 (splat.ply)} \rightarrow \textit{Browser}$$

The full pipeline runs as a serverless GPU function on Modal.com, spawned on demand and shut down automatically on completion. Live step progress is written to a

`status.json` file in Cloudflare R2 and polled by the frontend every 4 seconds.

B. Three-Tier Brightness Detection

Before any reconstruction work begins, the pipeline measures mean pixel brightness across all submitted images on a 0–255 scale. The result determines which processing path the submission follows, as summarised in Table II.

TABLE II
THREE-TIER ADAPTIVE BRIGHTNESS DETECTION AND ROUTING

Mean Brightness	Action	3D Output
$< 50/255$	Zero-DCE + Shadow Lift + YCbCr	No - enhanced photos only
$50–60/255$	Zero-DCE enhancement	Yes - continue to COLMAP
$60–100/255$	Gamma correction	Yes - continue to COLMAP
$\geq 100/255$	No enhancement	Yes - direct to COLMAP

C. Pipeline Stages

The full pipeline proceeds through five stages:

1) **Download** - The Modal worker downloads all accident photos from Cloudflare R2 at the claim prefix.
2) **Brightness check and enhancement** - Mean brightness is measured; the appropriate enhancement tier is applied.
3) **COLMAP** - Feature extraction, exhaustive matching, and sparse reconstruction to produce camera poses (skipped for critically dark inputs).
4) **nerfstudio splatfacto** - Gaussian Splat training using COLMAP camera poses on NVIDIA L4 GPU.
5) **Export and upload** - The trained splat is exported as `splat.ply` and uploaded to Cloudflare R2; `status.json` is marked complete.

D. AI and ML Components

Table III summarises the AI and ML components in the pipeline, distinguishing deep learning, classical computer vision, and custom signal processing stages.

E. Infrastructure and Cost

COLMAP and nerfstudio require a dedicated NVIDIA GPU with CUDA. Maintaining a GPU server continuously for an intermittently used system is economically unviable. Modal.com provides serverless NVIDIA L4 GPUs, the pipeline spawns a worker on demand, executes, and shuts down automatically, with billing per second of actual GPU usage. The estimated cost per 3D model generated is approximately \$0.50–\$1.50, with zero idle cost between submissions.

TABLE III
AI AND ML COMPONENTS IN THE PROPOSED PIPELINE

Stage	Technology	Type	Trained
Enhancement	Zero-DCE	Deep Learning	Pre-trained
Feature extraction	COLMAP/SIFT	Classical CV	None
Pose estimation	Bundle Adjustment	Math. Opt.	None
3D reconstruction	nerfstudio splatfacto	Neural Rendering	Per-scene
Shadow lift	Custom LUT	Signal Proc.	None
Noise reduction	YCbCr filters	Signal Proc.	None

F. Multi-Layer Fraud Verification Framework

A fundamental weakness of existing 2D-based claim systems is the inability to verify that submitted photographs were captured at the actual accident location and time. The proposed system anchors every claim to four independently recorded evidence points, as summarised in Table IV.

TABLE IV
MULTI-LAYER FRAUD VERIFICATION EVIDENCE POINTS

Evidence Point	Data Captured	Fraud Signal
Insurance call	Time, GPS, lat/lng	Location mismatches scene
Capture start	Time, GPS, lat/lng	Photos started away from scene
Per-image metadata	GPS per photo, timestamp	Images from different locations
Submission time	Time, GPS, lat/lng	User left scene before submitting

In Sri Lanka, vehicle owners must notify their insurer by telephone before any other action is taken. The system records the call time and GPS coordinates, the capture start time, per-image GPS metadata, and the submission timestamp. All four points must be spatially consistent any discrepancy raises an automatic fraud flag in the insurer dashboard. The system also requires the claimant to photograph their driving license, vehicle revenue license, and selfie with GPS metadata embedded, and for two-vehicle accidents, the same documents from the other driver.

The reconstructed 3D Gaussian Splat provides a passive spatial coherence check, physical objects at the accident scene, such as trees, barriers, or other vehicles, appear in the model. An insurance officer can visually confirm that reported damage corresponds to visible deformation in the reconstruction, making such inconsistencies more difficult to conceal and providing additional spatial evidence for claim verification.

IV. LOW-LIGHT ENHANCEMENT CHAIN

When mean brightness falls between 50 and 60/255, or below 50/255, the pipeline applies a three-stage enhancement chain before reconstruction or delivery.

A. Stage 1 - Zero-DCE Neural Enhancement

Zero-DCE [7] is a lightweight CNN (∼8 MB weights, `Epoch99.pth`) that predicts per-pixel tone curves without requiring paired bright/dark training data. The network applies 8 iterative curve passes, described by:

$$\hat{x}^{(n)} = \hat{x}^{(n-1)} + \alpha^{(n)} \cdot \hat{x}^{(n-1)} \cdot (1 - \hat{x}^{(n-1)}) \tag{1}$$

where $\hat{x}^{(n)}$ is the enhanced image at iteration n and $\alpha^{(n)}$ is the per-pixel curve parameter predicted by the network. Darker pixel regions receive larger α values, producing spatially-aware brightening that lifts shadows without overexposing existing highlights.

B. Stage 2 - Shadow Lift Lookup Table

Zero-DCE lifts overall image brightness, but shadow regions around tyres, wheel arches, and vehicle undersides can remain dark. A custom piecewise lookup table applies a targeted secondary lift:

$$f(p) = \begin{cases} p^{1/\gamma}, & p < 0.40 \quad (\gamma = 2.0) \\ \text{blend}(p^{1/\gamma},\ p), & 0.40 \leq p < 0.70 \\ p, & p \geq 0.70 \end{cases} \tag{2}$$

where p is normalised pixel brightness. Pixels below 40% brightness receive a strong gamma-2.0 lift; pixels above 70% are untouched, preserving highlights for reliable COLMAP feature matching.

C. Stage 3 - YCbCr Noise Reduction

Shadow lifting amplifies sensor noise previously hidden in dark regions. The image is converted from RGB to YCbCr colour space and treated per channel as summarised in Table V.

TABLE V
YCbCr Noise Reduction - Per-Channel Treatment

Channel	Filter	Rationale
Y (luminance)	MedianFilter (size=3)	Removes grain; preserves sharp edges
Cb (chroma)	GaussianBlur (radius=3)	Removes colour noise
Cr (chroma)	GaussianBlur (radius=3)	Removes colour noise

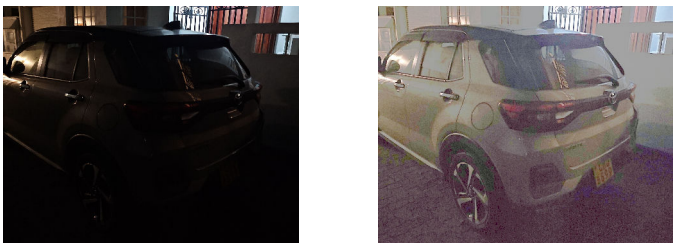


Fig. 1. Low-light accident photograph before (left) and after (right) the three-stage enhancement chain (mean brightness 33.0 → 123.1/255).

Image quality is measured using BRISQUE [11], a no-reference metric assessing perceptual quality without a clean reference image. A median filter is applied to the luminance channel because it cannot blur across hard edges, essential for preserving panel boundary features that COLMAP must detect. Gaussian blur is appropriate for chroma channels because the human visual system is significantly less sensitive to chroma resolution than luminance resolution.

V. Gaussian Splatting Reconstruction

A. Structure-from-Motion via COLMAP

COLMAP [5] recovers 3D scene geometry from unordered 2D photographs through three sequential steps. First, feature extraction uses SIFT [8] to identify distinctive keypoints corners, blobs, and edges that remain recognisable across different viewpoints. Second, exhaustive feature matching compares keypoints across all image pairs to identify the same physical point appearing in multiple photographs. Third, bundle adjustment [5], [12] solves a non-linear least squares problem to simultaneously optimise 3D point positions and camera poses, producing a sparse point cloud and a set of calibrated camera transformations.

COLMAP is classical computer vision, deterministic mathematical optimisation with no trained model. Its failure mode on dark images is well-defined: in shadow regions where pixel values approach zero, SIFT finds almost no distinctive features. A dark image set can produce as few as 2 feature matches across 9 photographs, far below the minimum required for sparse reconstruction. This is the technical justification for running brightness detection before COLMAP, not after.

B. Gaussian Splatting via nerfstudio

Gaussian Splatting [9] represents a 3D scene as a collection of millions of Gaussian primitives, each defined by a position $\boldsymbol{\mu} \in \mathbb{R}^3$, a covariance matrix $\boldsymbol{\Sigma}$ defining ellipsoid shape, a view-dependent colour $\mathbf{c}$ encoded as spherical harmonics, and an opacity $\alpha \in [0, 1]$. Training proceeds via gradient descent, minimising:

$$\mathcal{L} = (1 - \lambda)\mathcal{L}_1 + \lambda\mathcal{L}_{\text{D-SSIM}} \tag{3}$$

where $\mathcal{L}_1$ is a pixel-wise L1 loss and $\mathcal{L}_{\text{D-SSIM}}$ [13] is a structural similarity loss. A densification strategy adds new Gaussians in under-represented regions throughout training. The Poisson surface reconstruction algorithm [14] converts the dense point cloud into a watertight polygonal mesh during export. The nerfstudio splatfacto implementation [6] runs for 3,000 training steps on an NVIDIA L4 GPU. The trained model is exported as a `splat.ply` file and rendered in the browser using the `@mkkellogg/gaussian-splats-3d` WebGL library, enabling the insurance officer to rotate, pan, and zoom around the damaged vehicle with no plugins required.

VI. Experimental Results

A. Evaluation Setup

Over 30 real pipeline executions were conducted on using accident photographs submitted through the mobile application. Tables VI and VII report five runs covering well-lit and critically dark conditions across the defined brightness tiers in Table II. All executions ran on an NVIDIA L4 GPU provisioned via Modal.com.

B. Pipeline Timing Results

Table VI reports per-stage and total pipeline times for three representative executions.

TABLE VI
Pipeline Execution Times Across Five Real Runs

Stage	**R1** *43*	**R2** *39*	**R3** *36*	**R4** *39*	**R5** *42*
Download	26.5s	11.4s	—	22.1s	25.5s
Enhancement	34.1s	20.4s	68.0s	26.8s	26.6s
COLMAP	378.9s	701.1s	Skip	293.2s	325.1s
Training	711.5s	538.3s	Skip	604.6s	561.2s
Export	67.7s	71.1s	—	82.2s	67.3s
Total	**20.3m**	**22.4m**	**4m**	**17.2m**	**16.8m**

Run 3 completed in approximately 4 minutes because the critically dark condition (33.0/255) triggered intelligent degradation, COLMAP and nerfstudio training were skipped, and enhanced photographs were delivered instead. COLMAP dominated reconstruction time and exhibited significant variance: Runs 2 and 4 both processed 39 images yet took 701.1s and 293.2s respectively, confirming that mapper time is sensitive to scene complexity and feature distribution rather than image count alone.

C. Brightness and Enhancement Results

Table VII reports brightness and BRISQUE [11] quality metrics for all three runs.

TABLE VII
Brightness and BRISQUE Quality Metrics

Metric	**R1**	**R2**	**R3**	**R4**	**R5**
Images	43	39	36	39	42
Brightness (pre)	117.2	118.8	33.0	111.9	119.8
Brightness (post)	—	—	123.1	—	—
Brightness Δ	—	—	$+90.1$	—	—
BRISQUE (pre)	26.68	29.04	47.81	32.18	24.30
BRISQUE (post)	—	—	43.22	—	—
BRISQUE Δ	—	—	-4.59	—	—
Enhancement	None	None	Zero-DCE	None	None
3D generated	Yes	Yes	No	Yes	Yes

Runs 1, 2, 4, and 5 exceeded the well-lit threshold ($\geq 100/255$) and proceeded directly to COLMAP with no enhancement applied. Run 3 demonstrates the low-light path: the three-stage enhancement chain raised mean brightness from 33.0 to 123.1/255 ($+90.1$), with BRISQUE improving by 4.59 points, demonstrating visibility recovery with residual perceptual-quality degradation from shadow lifting. Gaussian Splat reconstruction quality across well-lit runs achieved PSNR 21.78 dB and SSIM 0.6536 [13], consistent with object-scale neural rendering benchmarks [9].



Fig. 2. Four viewpoints of a Gaussian Splat model generated by the proposed pipeline from a single accident photo submission.

D. Summary of Empirical Findings

Across over 30 real executions, the following baselines were observed: end-to-end reconstruction of 20.3–22.4 minutes on NVIDIA L4 GPU; nerfstudio throughput of 241–251 million rays/second; low-light enhancement in 68.0 seconds for 36 images; brightness restoration of $+90.1$ points ($33.0 \rightarrow 123.1/255$); BRISQUE improvement of $47.81 \rightarrow 43.22$ ($\Delta = -4.59$); correct intelligent degradation confirmed in Run 3; and reconstruction quality of PSNR 21.78 dB and SSIM 0.6536 [13].

VII. Limitations and Future Work

A. Reconstruction Quality

The nerfstudio splatfacto training achieved a final PSNR of 21.78 dB and SSIM [13] of 0.6536 on held-out evaluation views via TensorBoard logging, consistent with object-scale Gaussian Splat reconstructions from uncontrolled smartphone captures [9]. Future work will expand evaluation across multiple submissions to establish damage-severity-specific quality baselines.

B. COLMAP Mapper Time Variance

The observed variance in COLMAP mapper time (378.9s versus 701.1s for similar image counts) suggests sensitivity to scene geometry and feature distribution. Future work will investigate adaptive matching strategies - such as sequential or vocabulary-tree matching in place of exhaustive matching - to reduce mapper time variance for production use.

C. BRISQUE as Enhancement Metric

BRISQUE [11] penalises noise amplification in enhanced images regardless of their utility for COLMAP feature matching. A more appropriate metric is COLMAP feature match count before and after enhancement, which future work will instrument the pipeline to capture.

D. Dataset Scale

A larger controlled evaluation spanning minor, moderate, and structural damage categories is required to establish statistically robust reconstruction completeness baselines.

E. Reflective Surface Failure Mode

Gaussian Splatting degrades on highly reflective vehicle surfaces. Specular reflections cause COLMAP feature matching failures and introduce view-dependent colour inconsistencies during Gaussian optimisation. A polarised filter accessory for the mobile capture process is under consideration as a practical mitigation for metallic paint surfaces.

VIII. Conclusion

This paper presented a serverless Gaussian Splatting pipeline for photo-realistic 3D vehicle reconstruction in motor insurance claim processing. The system addresses three fundamental failures of existing 2D-based claim workflows through a five-stage pipeline integrating adaptive brightness detection, Zero-DCE neural enhancement, COLMAP Structure-from-Motion, and nerfstudio Gaussian Splatting on a serverless NVIDIA L4 GPU.

A key contribution is the three-tier brightness detection system and its associated intelligent degradation mechanism, which ensures the pipeline produces a useful output, either a photo-realistic 3D Gaussian Splat or enhanced photographs, across the evaluated lighting conditions.

Evaluated across over 30 real pipeline executions, with representative results showing end-to-end reconstruction times of 20.3 and 22.4 minutes for well-lit submissions and brightness restoration of +90.1 points for critically dark inputs.

Future work will configure nerfstudio PSNR/SSIM evaluation callbacks, instrument COLMAP for per-pair feature match counts, conduct a larger controlled evaluation spanning minor, moderate, and structural damage categories, investigate vocabulary-tree matching to reduce mapper time variance, and evaluate polarised filter accessories for reflective surface mitigation.

References

[1] K. Gupta, S. S. Prasad, and K. S. Venkatesh, "Advances in computer vision and AI for automotive insurance: a survey of damage assessment techniques," *Complex & Intelligent Systems*, vol. 11, pp. 1–24, 2024.

[2] N. Somnuk and K. Kanchanasut, "A 3D reconstruction of vehicle accidents from 2D images using a convolutional neural network," *Engineering and Applied Science Research*, vol. 48, no. 5, pp. 605–613, 2021.

[3] K. I. Kholodkov, A. A. Gukov, V. M. Scherbakov *et al.*, "3D reconstruction of an accident site," *Izvestiya, Atmospheric and Oceanic Physics*, vol. 60, pp. 963–969, 2024.

[4] G. Schrijver, D. K. Sarmah, and M. El-hajj, "Automobile insurance fraud detection using data mining: A systematic literature review," *Smart Cities and Society*, vol. 3, 2024.

[5] J. L. Schönberger and J. Frahm, "Structure-from-motion revisited," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016.

[6] M. Tancik, E. Weber, E. Ng, R. Li, B. Yi, T. Kerr, M. Turki, A. Milosh *et al.*, "Nerfstudio: A modular framework for neural radiance field development," in *ACM SIGGRAPH 2023 Conference Proceedings*, 2023.

[7] C. Guo, C. Li, J. Guo, C. C. Loy, J. Hou, S. Feng, and R. Chen, "Zero-reference deep curve estimation for low-light image enhancement," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2020.

[8] D. G. Lowe, "Distinctive image features from scale-invariant keypoints," *International Journal of Computer Vision*, vol. 60, no. 2, pp. 91–110, 2004.

[9] B. Kerbl, G. Kopanas, T. Leimkühler, and G. Drettakis, "3D Gaussian splatting for real-time radiance field rendering," in *ACM SIGGRAPH 2023 Conference Proceedings*, 2023.

[10] B. Mildenhall, P. P. Srinivasan, M. Tancik, J. T. Barron, R. Ramamoorthi, and R. Ng, "NeRF: Representing scenes as neural radiance fields for view synthesis," in *European Conference on Computer Vision (ECCV)*, 2020.

[11] A. Mittal, A. K. Moorthy, and A. C. Bovik, "No-reference image quality assessment in the spatial domain," *IEEE Transactions on Image Processing*, vol. 21, no. 12, pp. 4695–4708, 2012.

[12] B. Triggs, P. McLauchlan, R. Hartley, and A. Fitzgibbon, "Bundle adjustment: a modern synthesis," *Vision Algorithms: Theory and Practice*, pp. 298–372, 2000.

[13] Z. Wang, A. C. Bovik, H. R. Sheikh, and E. P. Simoncourt, "Image quality assessment: from error visibility to structural similarity," *IEEE Transactions on Image Processing*, vol. 13, no. 4, pp. 600–612, 2004.

[14] M. Kazhdan, M. Bolitho, and H. Hoppe, "Poisson surface reconstruction," in *Proceedings of the Fourth Eurographics Symposium on Geometry Processing*, 2006, pp. 61–70.